① 6.6 - 12

$JM \qquad\qquad MK$

$$J = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 6 & 0 & 0 \end{array}\right] \qquad K = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right]$$

$$M = \begin{bmatrix} m_{11} & m_{12} & \cdots \\ m_{21} & m_{22} & \cdots \\ \vdots & \vdots & \end{bmatrix}$$

$$\Rightarrow JM = \left[\begin{array}{cc|cc} m_{21} & m_{22} & m_{23} & m_{24} \\ 0 & 0 & 0 & 0 \\ \hline m_{41} & m_{42} & m_{43} & m_{44} \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$m_{22} = m_{11}$

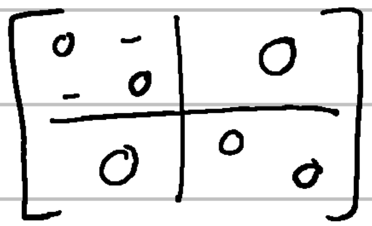
$$MK = \begin{bmatrix} 0 & m_{11} & m_{12} & 0 \\ 0 & m_{21} & m_{22} & 0 \\ 0 & m_{31} & m_{32} & 0 \\ 0 & m_{41} & m_{42} & 0 \end{bmatrix}$$

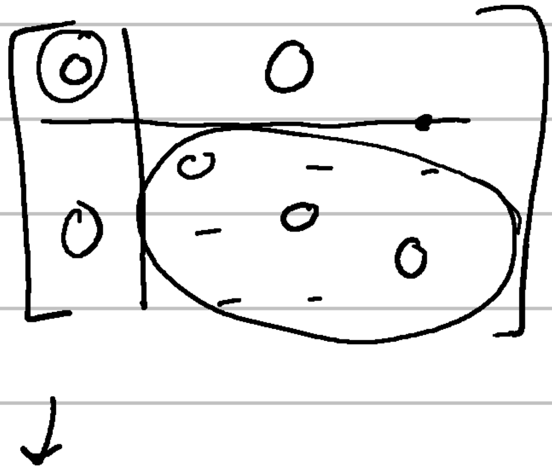
Hence not invertible

$$\begin{bmatrix} 0 & & \\ 0 & 0 & 0 \\ 0 & & \\ 0 & 0 & 0 \end{bmatrix}$$

② 6.6-13

5 Jordan forms





And 3 others like this

③ 6.6 – 14

To prove A^T is always similar to A

- We know that λ's are same

① J_i : Find M_i such that $M_i^{-1} J_i M_i = J_i^T$

⇒ **Neat fact** → J^T is always similar to J through reverse identity matrix

$$M = \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix} \qquad M^{-1} = \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix}$$

② $$M_0 = \begin{bmatrix} M_1^{-1} & & & \\ & M_2^{-1} & & \\ & & \ddots & \\ & & & M_n^{-1} \end{bmatrix}$$

③ $A = M J M^{-1}$

$$\underbrace{A^T = M^{-1} J^T}_{\text{similar}} M$$

$$\boxed{A^T \to J^T \to J \to A}$$

(4) G.G. 20

a) $B = M^{-1} A M$

$\underline{B^2} = M^{-1} \underline{A^2} M$

similar ✓ true

b) $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $B^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ $\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$$

d) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ is only similar to itself

$m^{-1} (3I) m = 3$

e) exchange row 1 & 2
then exchange column 1 & 2.

1^{st} op $\rightarrow$ $\begin{bmatrix} 0 & 1 & 0 & \cdots \\ 1 & 0 & 0 & \cdots \\ 1 & & & \end{bmatrix} A \begin{bmatrix} 0 & 1 & 0 & \cdots \\ 1 & 0 & \cdot & - \end{bmatrix}$

M

(8) 6.6 - 22

$A \to n \times n$ $\lambda = 0$

$$J = \begin{bmatrix} 0 & 0 & & & \\ & 0 & 0 & & \\ & & 0 & 0 & \\ & & & \ddots & \ddots \\ & & & & 0 \end{bmatrix}$$

$$J^2 = \begin{bmatrix} 0 & 0 & 0 & & \\ & 0 & 0 & 0 & \\ & & 0 & 0 & \ddots \\ & & & \ddots & 0 \\ & & & & 0 \end{bmatrix}$$

$$J^n = [0]$$

$$A = M^{-1} J M$$

$$A^n = M^{-1} J^n M = 0$$

⑥ 6.6 – 23

$A = QR$ $\quad\quad\quad\quad A_1 = RQ$

A_1 is similar to A because

$$\underbrace{RQ}_{A_1} = Q^{-1} \underbrace{(QR)}_{A} Q$$

→ Now in next step we do the same for $A_1 = Q_1 R_1 = RQ$

and then find $A_2 = R_1 Q_1$

Neat fact → This approach is the best way to find eigenvalues as A's approach a diagonal matrix.

$$A = QR$$

$$A_1 = RQ = Q_1 R_1 = Q^{-1} A Q$$

$$A_2 = R_1 Q_1 = Q_1^{-1} (\underbrace{Q_1 R_1}_{A_1}) Q_1$$

Hence A_2 is similar to A_1

$$\# \quad A_2 = Q_1^{-1} (Q^{-1} (QR) Q) Q_1$$

$$= (Q Q_1)^{-1} (QR) (Q Q_1)$$

① 6.6. 2a

$\Rightarrow$ If A is similar to A^{-1}

$$A = M^{-1} A^{-1} M$$

$\Rightarrow$ All eigen values are same for A & A^{-1}

$$\lambda_1 + \lambda_2 \ldots = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} \cdots$$

$$\lambda_1 \lambda_2 \cdots \lambda_n = \frac{1}{(\lambda_1 \cdots \lambda_n)}$$

$$\boxed{\begin{matrix} 1/2 \\ 2 \end{matrix}}$$

eg

$$\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

⑧ 6.7 - 4

$\Rightarrow \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \qquad \boxed{A = A^T}$

$$A^T A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A A^T = A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\lambda \rightarrow \quad (\lambda - 2)(\lambda - 1) - 1 = 0$$

$$\lambda^2 - 3\lambda + 1 = 0$$

$$\lambda = \frac{3 \pm \sqrt{9 - 4}}{2} = \frac{3 \pm \sqrt{5}}{2}$$

$$\lambda \rightarrow \frac{3 \pm \sqrt{5}}{2}$$

$$\boxed{\sqrt{\lambda} = \frac{1 \pm \sqrt{5}}{2}}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{4 - (3 + \sqrt{5})}{2} & 1 \\ 1 & \frac{2 - (3 + \sqrt{5})}{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{1 - \sqrt{5}}{2} & 1 \\ 1 & \frac{-\sqrt{5} - 1}{2} \end{bmatrix}$$

$$\frac{(-\sqrt{5} - 1)}{2} - \frac{2}{1 - \sqrt{5}}$$

$$\frac{-(1+\sqrt{5})(1-\sqrt{5}) - 4}{}$$

$$\frac{2}{1 - \sqrt{5}}$$

$$= -(1 - 5) - 4$$

$$= \underline{\underline{0}}$$

$$\frac{(1-\sqrt{5})}{2} x = -1$$

$$\sqrt{\frac{5+1-2\sqrt{5}+4}{\sqrt{5}-1}} = \sqrt{\frac{(10-2\sqrt{5})}{\sqrt{5}-1}}$$

$$x = \frac{2}{\sqrt{5}-1}$$

$$= \sqrt{\frac{2\sqrt{5}}{\sqrt{5}-1}}$$

$$x_1 = \begin{bmatrix} \frac{2}{\sqrt{5}-1} \\ 1 \end{bmatrix}$$

$$q_1 = \sqrt{\frac{4}{\sqrt{5}-1} 2 \times \frac{}{2\sqrt{5}}}$$

$$q_1 = \sqrt{\frac{2}{5-\sqrt{5}}}$$

$$x_1 = \begin{bmatrix} \sqrt{\frac{2}{5-\sqrt{5}}} \\ \sqrt{\frac{\sqrt{5}-1}{2\sqrt{5}}} \end{bmatrix}$$

we can just use eigen decomposition

$$A = S \Lambda S^{-1}$$

$$S \downarrow U \qquad S^{-1} \downarrow V^T$$

9. 6.7 - 11

$A \to$ ortho colums $w_1, w_2 \ldots$

$|w_1| = \sigma_1$

$|w_2| = \sigma_2$

$\vdots$

$A^T \Rightarrow$ ortho

$\Rightarrow \quad A = U \Sigma V^T$

$A = U \Sigma V^T$

$$A^T A = \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \end{bmatrix}$$

$A^T A = V \Sigma^2 V^T$

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}$$

$A A^T = U \Sigma^2 U^T$

$U \to$ orthonormalized A

$\boxed{V = I}$

$$U = \begin{bmatrix} | & | & \\ w_1/\sigma_1 & w_2/\sigma_2 & \cdots \\ | & | & \end{bmatrix}$$

(10) 6.7-17

$A = U \Sigma V^T$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \qquad A^TA = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} \sin \pi/4 & \sin 2\pi/4 & \sin 3\pi/4 \\ \sin 2\pi/4 & \sin 4\pi/4 & \sin 6\pi/4 \\ \sin 3\pi/4 & \sin 6\pi/4 & \sin 9\pi/4 \end{bmatrix}$$

a)

$$\begin{bmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{bmatrix}$$

$$(2-\lambda)\left((2-\lambda)^2 - 1\right) - (2-\lambda) = 0$$

$$(2-\lambda)(\lambda^2 + 4 - 2\lambda - 2) = 0$$

$$(\lambda-2)(\lambda^2 - 2\lambda + 2) = 0$$

$$\lambda = 2-\sqrt{2}, \quad 2, \quad 2+\sqrt{2}$$

$$x_1 \rightarrow \begin{bmatrix} \sqrt{2} & -1 & 0 \\ -1 & \sqrt{2} & -1 \\ 0 & -1 & \sqrt{2} \end{bmatrix} \qquad \begin{bmatrix} 1/2 \\ 1/\sqrt{2} \\ 1/2 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{2} & -1 & 0 \\ 0 & 1/\sqrt{2} & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

1st col of V

$$x_2 \rightarrow \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$$

↳ 2nd col

$$x_3 \rightarrow \begin{bmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{bmatrix}$$

$$x_3 \rightarrow \begin{bmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -\sqrt{2} & -1 & 0 \\ 0 & -1/\sqrt{2} & -1 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1/2 \\ -1/\sqrt{2} \\ 1/2 \end{bmatrix}$$

3rd colm

$$-\sqrt{2}x + \sqrt{2} = 0$$

$$x = 1$$

b) Multiply AV

$$AV = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 1/2 & 1/2 & 1/2 \\ \frac{\sqrt{2}-1}{2} & -\frac{1}{2} & -\frac{(1+\sqrt{2})}{2} \\ \frac{1-\sqrt{2}}{2} & -1/2 & \frac{\sqrt{2}+1}{2} \\ -1/2 & 1/2 & -1/2 \end{bmatrix}$$

$\boxed{\text{Ortho} \checkmark}$

c) Nullspace of A^T

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$N(A^T) = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

(11) $8.5-4$

$1, \quad x, \quad x^2 - 1/3$

$\Rightarrow (x^3 - cx)(x)$

$$\int_{-1}^{1} x^4 - cx^2,$$

$$\left. \frac{x^5}{5} - \frac{cx^3}{3} \right]_{-1}^{1}$$

$$2\left(\frac{1}{5} - \frac{c}{3}\right) = 0$$

$$\frac{c}{3} = \frac{1}{5} \qquad c = 3/5$$

(12) 8.5-5

$$f(x) = \frac{4}{\pi}\left[\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} \ldots\ldots\right.$$

$$\int_0^{2\pi} f(x) \cos x \, dx$$

$\sin x$

$\sin 2x$

Need to study integrals but the coeff would be the ones corresponding to

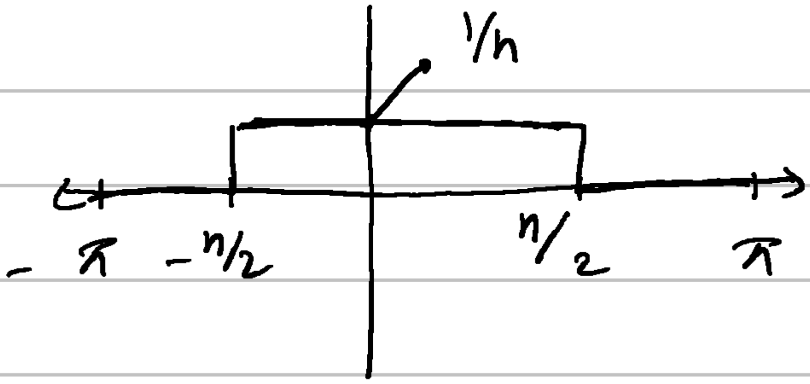
$\cos x$, $\sin x$ & $\sin 2x$ terms respectively

(13) 3.5 — 12

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ \cos x \\ \sin x \\ \cos 2x \\ \sin 2x \end{bmatrix} = \begin{bmatrix} 0 \\ -\sin x \\ \cos x \\ -2\cdot\sin 2x \\ 2\cdot\cos 2x \end{bmatrix}$$

(14) 8.5 - 13

$F(x) \rightarrow$



$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\,dx = \frac{1}{h\pi}\, x\Big]_{-h/2}^{h/2} = \frac{1}{\pi}$$

$$b_i = \int_{-h/2}^{h/2} \sin(nx)\,dx = 0$$

$$a_i = \frac{1}{\pi k h}\int_{-h/2}^{h/2} \cos(kx) = \frac{2}{\pi h k}\sin\frac{kh}{2}$$

Note: need to study integrals

$\lim h \to 0$, $a_0 = \frac{1}{\pi}$